

Health and Physical Education

A practical wellness handbook covering safe exercise, flexibility, yoga, weight training, sports, games, teamwork and lifelong healthy habits.

Course 1009

Semester I

Practicum

Scheme 0:0:2:0

Credits 0

Official Source Declaration

The handbook is grounded in the verified Revision 2026 SITTTTR extracted source for Course 1009 — Health and Physical Education. It preserves the practical emphasis on health awareness, warm-up and cool-down, flexibility, yoga and pranayama, weight training, sports, games, teamwork, discipline and fitness.

The official syllabus source is authoritative for the course code, title, scheme, outcomes, modules, hours, activities, assessment and references. Any source-format irregularity must remain visible to the learner and examiner.

Course Dashboard

Course information

Field	Official value
Course code	1009
Course title	Health and Physical Education
Semester	I
Course type	Practicum
Teaching scheme	0:0:2:0
Credits	0
Module hours	As shown in the official module coverage table
Lesson purpose	Handbook-scale study, practical preparation and examination revision

CO	Outcome	Bloom level
CO1	Introduction to health and physical education; perform warming-up and warming-down exercises.	Apply
CO2	Apply basic physical training methods to improve stretching and flexibility exercises correctly.	Apply
CO3	Demonstrate selected yoga asanas and pranayama techniques for wellness.	Apply
CO4	Participate effectively in sports and games; demonstrate teamwork, discipline and muscular-strength development.	Apply

CO-PO correlation matrix

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	2	2	2	2	-	2	-
CO2	-	-	-	-	-	2	2	2	2	-	2	-
CO3	-	-	-	-	-	3	3	2	2	-	3	-
CO4	-	-	-	-	-	3	3	3	2	-	3	-
Total	-	-	-	-	-	10	10	9	8	-	10	-
Correlation Level	-	-	-	-	-	2.5	2.5	2.25	2	-	2.5	-

Complete Syllabus Coverage

Official module coverage

Module	Official entries	Hours
I — Introduction and warm-up	Introduction to Health and Physical Education; Warm-up and Cool-down Exercises	2
II — Flexibility	Flexibility Exercises — Basics; Practice; Benefits	4
III — Yoga	Yoga Asanas and Pranayama	4
IV — Weight training, health, sports and games	Weight Training and Health Awareness; Sports and games; Open-ended experiment	20
Total	Practical entries plus open-ended experiment	30

Learning Roadmap



Use the roadmap to connect individual practical tasks to course outcomes.



Practical competence grows through preparation, performance, feedback and reflection.

MODULE 1

Health Education and Warm-up

Official hours: 2 · CO1

Health education, warm-up and cool-down as the safe foundation for activity.

Learning goals

- Introduction to health and physical education
- Warm-up and cool-down exercises

Key facts

- Warm-up is progressive preparation and cool-down is gradual recovery.
- Posture and breathing influence safe movement.
- Stretching must be controlled and progressive.

- Teamwork, rules and discipline are practical skills.

Engineering relevance: Safe physical practice supports attention, resilience, cooperation and healthy routines in engineering life.

Introduction to health and physical education

Explain meaning, aims and objectives; connect physical health, mental well-being, social development and safety.

Malayalam support: Safety ഇപ്പോൾ instructor guidance ഇപ്പോൾ. Pain, dizziness, unusual breathlessness ഇപ്പോൾ activity ഇപ്പോൾ report ഇപ്പോൾ.

Safety: This handbook is educational. It does not diagnose illness or replace qualified medical advice.

Warm-up and cool-down exercises

Check space and hydration; progress from low intensity; use dynamic preparation; reduce intensity gradually; report pain or unusual symptoms.

Malayalam support: Safety ഇപ്പോൾ instructor guidance ഇപ്പോൾ. Pain, dizziness, unusual breathlessness ഇപ്പോൾ activity ഇപ്പോൾ report ഇപ്പോൾ.

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MODULE 2

Posture, Flexibility and Stretching

Official hours: 4 • CO2

Learning goals

- Flexibility exercises — basics
- Flexibility exercises — practice
- Flexibility exercises — benefits

Key facts

- Warm-up is progressive preparation and cool-down is gradual recovery.
- Posture and breathing influence safe movement.
- Stretching must be controlled and progressive.
- Teamwork, rules and discipline are practical skills.

Engineering relevance: Safe physical practice supports attention, resilience, cooperation and healthy routines in engineering life.

Flexibility exercises — basics

Describe posture types and identify major bones and muscles involved in physical activity.

Malayalam support: Safety ഇൻസ്ട്രക്ടർ ഗൈഡൻസ് ഉപയോഗിച്ച്. Pain, dizziness, unusual breathlessness എന്നിവ ഉണ്ടായാൽ activity നിലയ്ക്കിട്ട് report ചെയ്യുക.

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Flexibility exercises — practice

Perform stretching correctly: warm up, move into comfortable range, breathe normally, avoid forceful bouncing and record observations.

Malayalam support: Safety☐☐☐ instructor guidance☐☐ ☐☐☐☐☐. Pain, dizziness, unusual breathlessness ☐☐☐☐☐☐ ☐☐☐☐☐☐ activity ☐☐☐☐☐☐☐ report ☐☐☐☐☐☐☐.

Safety: This handbook is educational. It does not diagnose illness or replace qualified medical advice.

Flexibility exercises — benefits

Compare dynamic, static, active and assisted stretching and explain injury-prevention relevance without overclaiming.

Malayalam support: Safety☐☐☐ instructor guidance☐☐ ☐☐☐☐☐. Pain, dizziness, unusual breathlessness ☐☐☐☐☐☐ ☐☐☐☐☐☐ activity ☐☐☐☐☐☐☐ report ☐☐☐☐☐☐☐.

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MODULE 3

Yoga Asanas and Pranayama

Official hours: 4 · CO3

Selected postures and breathing practices for wellness and disciplined practice.

Learning goals

- Yoga asanas and pranayama

Key facts

- Warm-up is progressive preparation and cool-down is gradual recovery.
- Posture and breathing influence safe movement.
- Stretching must be controlled and progressive.
- Teamwork, rules and discipline are practical skills.

Engineering relevance: Safe physical practice supports attention, resilience, cooperation and healthy routines in engineering life.

Yoga asanas and pranayama

Prepare a clean non-slip area; enter and leave postures slowly; use comfortable breathing; explain preventive wellness benefits carefully.

Malayalam support: Safety ഇന്ദ്രിയം instructor guidance ശാരീരികം. Pain, dizziness, unusual breathlessness ശാരീരികം ശാരീരികം activity ശാരീരികം report ശാരീരികം.

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MODULE 4

Weight Training, Health Awareness, Sports and Games

Official hours: 20 · CO4

Fitness, health awareness, weight training, sports, games, leadership and teamwork.

Learning goals

- Health-risk behaviours, hypokinetic disease and nutrition
- Weight training and musculoskeletal effects
- Sports and games — rules and fundamental skills
- Sports and games — mental and social skills
- Sports and games — leadership and teamwork

Key facts

- Warm-up is progressive preparation and cool-down is gradual recovery.
- Posture and breathing influence safe movement.
- Stretching must be controlled and progressive.
- Teamwork, rules and discipline are practical skills.

Engineering relevance: Safe physical practice supports attention, resilience, cooperation and healthy routines in engineering life.

Health-risk behaviours, hypokinetic disease and nutrition

Analyse inactivity-related risk concepts, nutrition, hydration, recovery and social well-being; do not self-diagnose.

Malayalam support: Safety ഇൻസ്ട്രക്ടർ ഗൈഡൻസ് ഉപയോഗിക്കുക. Pain, dizziness, unusual breathlessness ഉണ്ടായാൽ activity തടയുകയും report ചെയ്യുക.

Safety: This handbook is educational. It does not diagnose illness or replace qualified medical advice.

Weight training and musculoskeletal effects

Use safe load selection, neutral posture, controlled range, gradual progression, correct breathing and recovery.

Malayalam support: Safety☐☐☐ instructor guidance☐☐ ☐☐☐☐☐. Pain, dizziness, unusual breathlessness ☐☐☐☐☐☐ ☐☐☐☐☐☐ activity ☐☐☐☐☐☐☐ report ☐☐☐☐☐☐☐.

Safety: This handbook is educational. It does not diagnose illness or replace qualified medical advice.

Sports and games — rules and fundamental skills

Identify playing area, equipment and rules; practise skills progressively; use fair play and respect officials.

Malayalam support: Safety☐☐☐ instructor guidance☐☐ ☐☐☐☐☐. Pain, dizziness, unusual breathlessness ☐☐☐☐☐☐ ☐☐☐☐☐☐ activity ☐☐☐☐☐☐☐ report ☐☐☐☐☐☐☐.

Safety: This handbook is educational. It does not diagnose illness or replace qualified medical advice.

Sports and games – mental and social skills

Transfer coordination, communication, stress control and problem solving from sports to real-life tasks.

Malayalam support: Safety ് instructor guidance ് . Pain, dizziness, unusual breathlessness ് activity ് report ്.

Safety: This handbook is educational. It does not diagnose illness or replace qualified medical advice.

Sports and games — leadership and teamwork

Demonstrate punctuality, cooperation, fairness, unity, tolerance, inclusive roles and constructive feedback.

Malayalam support: Safety instructor guidance . Pain, dizziness, unusual breathlessness activity report .

Safety: This handbook is educational. It does not diagnose illness or replace qualified medical advice.

Formula, Procedure and Safety Bank

Practical record pattern

1. Aim
2. Requirements
3. Safety precautions
4. Procedure
5. Observation
6. Result

7. Reflection

Given: A selected activity and a supervised practical session.

Required: A safe, repeatable activity record.

Calculation / procedure: Write the planned sequence, perform under supervision, record observations and compare with the expected result.

Answer / expected result: The record shows safe procedure, participation and reflection.

Diagram Library for Rapid Revision

[Learning roadmap](#)

Foundation → practice → evidence → reflection.

[Module diagrams](#)

Each module contains a visual or structured checklist where it improves understanding.

The roadmap and safety loop diagrams support rapid revision. Use the practical record pattern for every supervised activity.

Official Self-learning Activities

Official self-learning activities

Week	Module	Activity	Hours
1	I	Recall and describe warm-up and cool-down concepts.	2
2	I	Perform warm-up exercises practised in class and demonstrate correct technique.	2
3	I	Explain the WHO definition of health after an online search.	2
4	I	Complete a test on basic health and physical education concepts.	2
5	II	Participate in and demonstrate known minor games.	2

Week	Module	Activity	Hours
6	II	Explain benefits of minor games.	2
7	II	Identify and summarise benefits of yoga using online resources.	2
8	II	Complete a test on games and yoga.	2
9	III	Define and explain Asana using online sources.	2
10	III	Prepare and present an assignment on minor games.	2
11	III	Perform asanas learned in class with correct posture.	2
12	III	Complete a test on yoga practices.	2
13	IV	Identify and explain major hypokinetic diseases through web research.	2
14	IV	Recall and report the number of bones in the human body using online resources.	2
15	IV	Explain different weight-training methods based on web research.	2

Assessment Methodology

Practical assessment methodology

Component	Timing	Marks
Continuous evaluation	Every practical slot	50 converted to 30
CA1 practical test	7th week; 2 hours	40 converted to 15
CA2 practical test	14th week; 2 hours	40 converted to 15
Open-ended experiment	15th week; 3 hours	50 converted to 10
ESE practical	3 hours	40
Total	—	CIE 60 + ESE 40 = 100

Practical test/ESE evaluation

Category	Marks
Yoga/Aerobics	10
Physical fitness test	10
Viva voce	10
Record	10

Category	Marks
Total	40

Malayalam support: Activity പരീക്ഷാപരിപാടി preparation, participation, discipline, teamwork, safety, record, viva പരീക്ഷാ പരീക്ഷാപരിപാടി.

Module-wise Questions and Answers

What is warm-up?

Warm-up is gradual preparation before more demanding activity. It supports movement readiness and safer transition.

What is an asana?

An asana is a controlled yoga posture performed with alignment and comfortable breathing.

State two stretching precautions.

Warm up, avoid forceful bouncing, breathe normally, move through a comfortable range and stop for sharp pain.

Why are rules important in sports?

Rules support fairness, safety, role clarity and respectful participation.

How can sports skills help engineering teamwork?

Coordination, communication, punctuality, stress control and problem solving transfer to technical team tasks.

Practice Model Paper — Not an Official Examination Paper

Instructions: Answer all short questions and attempt the practical tasks using the official record format. This model is for practice only.

1. Define one key term from Module I and explain one safety precaution.
2. Describe a complete procedure from Module II with observations and expected result.
3. Compare two methods or tools from Module III.
4. Write a short reflection showing teamwork, discipline or ethical digital practice from Module IV.
5. Design a practical record page containing aim, requirements, procedure, observation, result and reflection.

Quick Revision

Module checklist

- Health meaning and warm-up/cool-down.
- Posture, flexibility, stretching types and benefits.
- Asanas, pranayama, alignment and safety.
- Nutrition, weight training, sports rules, teamwork and leadership.

Exam checklist

- Define terms accurately.

- Write procedure in sequence.
- Mention safety and supervision.
- Distinguish educational explanation from medical diagnosis.
- Record practical evidence neatly.

Glossary

Important technical terms

Term	Short meaning
Health education	Learning that supports physical, mental and social well-being.
Warm-up	Gradual preparation before more demanding activity.
Cool-down	Gradual reduction of activity after exercise.
Flexibility	Controlled available range of movement.
Posture	Alignment and arrangement of the body.
Asana	A controlled yoga posture.
Pranayama	Regulated breathing practice.
Hypokinetic disease	A study term for health risks associated with inadequate movement; not a self-diagnosis.
Teamwork	Coordinated participation toward a shared goal.

Official References and Resources

Official references

Reference
Health and Physical Education — Puri, K. and Chandra — Surjeet Publication.
Creating a Life of Health and Fitness — Dr Jerrold S. Greenberg and George B. Dintiman — Benjamin Cummings / Allyn & Bacon.
Fit and Well — Fashey, Tomas D., Insel, Paul M. and Roth, Walton T. — McGraw-Hill.
Official web resources listed in the syllabus: bodybuilding.com , livestrong.com/get-fit , yogabasics.com , exrn.net/WeightTraining and sportsknowhow.com/rules/index.html .

Equipment and software

Category	Official requirement
Major equipment	NIL
Major software	NIL

In-page Search